

Normal Distribution [continued]

Recall.

i) $\underset{\sim}{X} \sim N_p(\underset{\sim}{\mu}, \underset{\sim}{\Sigma})$; $\underset{\sim}{\mu} \in \mathbb{R}^p$ and $\underset{\sim}{\Sigma}$ is p.s.d. ($\underset{\sim}{\Sigma} \geq 0$)

iff for any $\underset{\sim}{a} \in \mathbb{R}^p$; $\underset{\sim}{a}^T \underset{\sim}{X} \sim N(\underset{\sim}{a}^T \underset{\sim}{\mu}, \underset{\sim}{a}^T \underset{\sim}{\Sigma} \underset{\sim}{a})$

ii) If $\underset{\sim}{X} \sim N(\underset{\sim}{\mu}, \underset{\sim}{\Sigma}) \iff \phi_{\underset{\sim}{X}}(\underset{\sim}{t}) = E(e^{\underset{\sim}{i} \underset{\sim}{t}^T \underset{\sim}{X}}) = \exp \left\{ \underset{\sim}{i} \underset{\sim}{\mu}^T \underset{\sim}{t} - \frac{1}{2} \underset{\sim}{t}^T \underset{\sim}{\Sigma} \underset{\sim}{t} \right\}$

[1] $\underset{\sim}{X} \sim N_p(\underset{\sim}{\mu}, \underset{\sim}{\Sigma})$. Then

$$\underset{\underset{q \times 1}{\sim}}{Y} = \underset{\underset{q \times p}{\sim}}{A} \underset{\sim}{X} + \underset{\sim}{b} \sim N_q(\underset{\sim}{A} \underset{\sim}{\mu} + \underset{\sim}{b}, \underset{\underset{q \times q}{\sim}}{A} \underset{\sim}{\Sigma} \underset{\sim}{A}^T)$$

Proof:

$$\begin{aligned} \phi_{\underset{\sim}{Y}}(\underset{\sim}{t}) &= E \left(e^{\underset{\sim}{i} \underset{\sim}{t}^T \underset{\sim}{Y}} \right) = E \left(e^{\underset{\sim}{i} \underset{\sim}{t}^T (\underset{\sim}{A} \underset{\sim}{X} + \underset{\sim}{b})} \right) \\ &= E \left(e^{\underset{\sim}{i} \underset{\sim}{t}^T \underset{\sim}{A} \underset{\sim}{X}} \underline{e^{\underset{\sim}{i} \underset{\sim}{t}^T \underset{\sim}{b}}}} \right) \\ &= e^{\underset{\sim}{i} \underset{\sim}{t}^T \underset{\sim}{b}} \phi_{\underset{\sim}{X}}(\underset{\sim}{A}^T \underset{\sim}{t}) \\ &= e^{\underset{\sim}{i} \underset{\sim}{t}^T \underset{\sim}{b}} e^{\underset{\sim}{i} \underset{\sim}{t}^T \underset{\sim}{A} \underset{\sim}{\mu} - \frac{1}{2} \underset{\sim}{t}^T \underset{\sim}{A} \underset{\sim}{\Sigma} \underset{\sim}{A}^T \underset{\sim}{t}} \end{aligned}$$

Matching with the characteristic function of normal distribution, by uniqueness of char fn., we get $\underset{\sim}{Y} \sim N_q(\underset{\sim}{A} \underset{\sim}{\mu} + \underset{\sim}{b}, \underset{\sim}{A} \underset{\sim}{\Sigma} \underset{\sim}{A}^T)$.

Note: $\underset{\sim}{A}$ can be any rectangular matrix.

Example: Joint of $\underset{\sim}{Y} = \underset{\sim}{A} \underset{\sim}{X} + \underset{\sim}{b}$ and $\underset{\sim}{X}$

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$\underset{\sim}{X} \sim N(\underset{\sim}{\mu}, \underset{\sim}{\Sigma})$; $\underset{\sim}{Y} = \underset{\sim}{A} \underset{\sim}{X} + \underset{\sim}{b}$.

$$\underset{\sim}{Z} = \begin{bmatrix} \underset{\sim}{Y} \\ \underset{\sim}{X} \end{bmatrix} \quad \underset{\sim}{Z} = \underset{\sim}{C} \underset{\sim}{X} + \underset{\sim}{D} \quad \underset{\sim}{C} = \begin{bmatrix} \underset{\sim}{A} \\ \underset{\sim}{I}_p \end{bmatrix}_{(q+p) \times p} \quad \underset{\sim}{D} = \begin{bmatrix} \underset{\sim}{b} \\ 0 \end{bmatrix}$$

$$\underset{\sim}{C} \underset{\sim}{\Sigma} \underset{\sim}{C}^T = \begin{bmatrix} \underset{\sim}{A} \underset{\sim}{\Sigma} \\ \underset{\sim}{\Sigma} \end{bmatrix} \begin{bmatrix} \underset{\sim}{A}^T & \underset{\sim}{I}_p \end{bmatrix} \quad \underset{\sim}{Z} \sim N_{q+p} \left(\begin{bmatrix} \underset{\sim}{A} \underset{\sim}{\mu} + \underset{\sim}{b} \\ \underset{\sim}{\mu} \end{bmatrix}, \begin{bmatrix} \underset{\sim}{A} \underset{\sim}{\Sigma} \underset{\sim}{A}^T & \underset{\sim}{A} \underset{\sim}{\Sigma} \\ \underset{\sim}{\Sigma} \underset{\sim}{A}^T & \underset{\sim}{\Sigma} \end{bmatrix} \right)$$

[2] If $X \sim N_p(\mu, \Sigma)$ then $E(X) = \mu$, $\text{Var}(X) = \Sigma$.

Proof $E(X) = \begin{bmatrix} E(X_1) \\ \vdots \\ E(X_p) \end{bmatrix}$; $\text{Var}(X) = ((\text{Cov}(X_i, X_j)))$

$$X_i = e_{\sim_i}^T X \sim N(\mu_i, \sigma_{ii}) \text{ where } e_{\sim_i} = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \leftarrow i\text{th}$$

$$E(X_i) = \mu_i ; \text{var}(X_i) = \sigma_{ii} \text{ --- } \textcircled{1}$$

$$E(X) = \mu. \text{ --- } \textcircled{1}$$

$$\mu = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_p \end{bmatrix}, \quad \Sigma = ((\sigma_{ij}))$$

$$(e_{\sim_i} + e_{\sim_j})^T X = X_i + X_j \sim N(\mu_i + \mu_j, \sigma_{ii} + \sigma_{jj} + 2\sigma_{ij})$$

$$E(X_i + X_j) = \mu_i + \mu_j$$

$$\text{Var}(X_i + X_j) = \sigma_{ii} + \sigma_{jj} + 2\sigma_{ij} = \underbrace{\text{var}(X_i)}_{\sigma_{ii}} + \underbrace{\text{var}(X_j)}_{\sigma_{jj}} + 2\text{Cov}(X_i, X_j)$$

$$\text{Cov}(X_i, X_j) = \sigma_{ij} \text{ --- } \textcircled{2}$$

$$\therefore \text{Var}(X) = ((\sigma_{ij})) = \Sigma.$$

[3] $X \sim N(\mu, \Sigma); \Sigma \geq 0; \Sigma = BB^T$

$$\iff X \stackrel{D}{\sim} BZ + \mu \quad (X \stackrel{D}{\sim} Y \text{ means } X \text{ and } Y \text{ have same distn})$$

where $Z \sim N_p(0, I)$

$$\Sigma \geq 0. \Sigma = U\Lambda U^T, U^T U = I_p, B = U\Lambda^{1/2}, UU^T = I_p$$

Proof: $[\Sigma > 0]$ $\Sigma = U\Lambda U^T$ with $UU^T = U^T U = I_{p \times p}$.

$$\text{Define } \Lambda^{1/2} = \text{Diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_p})$$

$$B = U\Lambda^{1/2}$$

$$\Lambda = \text{Diag}(\lambda_1, \dots, \lambda_p) \lambda_j > 0 \forall j \text{ as } \Sigma > 0.$$

$$\text{Also symmetric root: } B = U\Lambda^{1/2}U^T, B^{-1} = U\Lambda^{-1/2}U^T \text{ where } \Lambda^{-1/2} = \text{Diag}(\frac{1}{\sqrt{\lambda_1}}, \dots, \frac{1}{\sqrt{\lambda_p}})$$

$$(\Leftarrow \text{ part}) Z \sim N_p(0, I)$$

$$BZ + \mu \sim N_p(\mu, BB^T = \Sigma) \quad X \stackrel{D}{\sim} BZ + \mu$$

$$(\Rightarrow \text{ part}) X \sim N_p(\mu, \Sigma)$$

$$Z = B^{-1}(X - \mu) \sim N_p(0, I) \implies X \stackrel{D}{\sim} BZ + \mu$$

[4]

$$\begin{aligned} \tilde{X} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}_{p_2}^{p_1} &\sim N\left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}\right) \quad p_1 + p_2 = p \\ X_1 \perp\!\!\!\perp X_2 &\iff \Sigma_{12} = 0 \end{aligned}$$

Proof: We will use the result

$$\phi\left(\begin{matrix} X_1 \\ X_2 \end{matrix}\right)\left(\begin{pmatrix} t_1 \\ t_2 \end{pmatrix}\right) = \phi_{X_1}(t_1)\phi_{X_2}(t_2) \iff X_1 \perp\!\!\!\perp X_2$$

$$\phi_{\tilde{X}}(t) = \dots = \phi_{X_1}(t_1)\phi_{X_2}(t_2) \quad t = \begin{bmatrix} t_1 \\ t_2 \end{bmatrix} \quad \text{iff } \Sigma_{12} = 0.$$

[5] **(Density)** $\tilde{X} \sim N(\mu, \Sigma)$; $\Sigma > 0$ then density of $\tilde{X}$ exists and is of the form $f_{\tilde{X}}(x)$:

Proof: $B = U\Lambda^{1/2}U^T$

$$\Sigma = U\Lambda U^T$$

$$B^{-1} = U\Lambda^{-1/2}U^T$$

$$\tilde{Z} = B^{-1}(\tilde{X} - \mu) \sim N(0, I_p)$$

$Z_1, \dots, Z_p$ are iid $N(0, 1)$ distn.

$$f_{\tilde{Z}}(z) = (2\pi)^{-p/2} \exp\left\{-\frac{1}{2} \sum_{i=1}^p z_i^2\right\}$$

$$\tilde{X} = B\tilde{Z} + \mu,$$

$$J_{z \rightarrow x} = \left|\det\left(\frac{\partial z}{\partial x}\right)\right| = \|B^{-1}\| = \frac{1}{\|B\|} = \frac{1}{|B|}$$

$$\sum_{i=1}^p z_i^2 = \tilde{Z}^T \tilde{Z} = (x - \mu)^T (B^{-1})^T B^{-1} (x - \mu) = (x - \mu)^T (BB^T)^{-1} (x - \mu) = (x - \mu)^T \Sigma^{-1} (x - \mu)$$

$$f_{\tilde{X}}(x) = f_{\tilde{Z}}(z) |J_{z \rightarrow x}|_{z=B^{-1}(x-\mu)} = (2\pi)^{-p/2} \exp\left\{-\frac{1}{2}(x - \mu)^T \Sigma^{-1} (x - \mu)\right\} \frac{1}{\|B\|}$$

$$|\Sigma| = |BB^T| = |B|^2 \implies \frac{1}{\|B\|} = \frac{1}{|\Sigma|^{1/2}}$$

Equi-probability contours

$$P(q) = \{x : f_X(x) \geq q\}$$

$$= \{x : (2\pi)^{-p/2} |\Sigma|^{-1/2} \exp\left[-\frac{1}{2}(x - \mu)^T \Sigma^{-1} (x - \mu)\right] \geq q\}$$

$$= \{x : (x - \mu)^T \Sigma^{-1} (x - \mu) \leq -2 \log q - p \log(2\pi) - \log |\Sigma|\}$$

$$q = 0.25 \quad q' = 0.50 \quad q = 0.95$$

